7.1 Type Conversion in C

Type conversion occurs when values of different data types are used in an expression. It can happen automatically through implicit conversion or manually through explicit casting.

7.2 Integer Division

When two integers are divided, the fractional part is discarded. For example, "150 / 100" produces "1" instead of "1.5". Using floating-point values or integer paise appropriately prevents this problem.

7.3 Explicit Casting

Explicit casting converts a value from one data type to another. For example, "(int)3.99" produces "3" because the fractional part is discarded. Casting a large "long long" value to "int" may also lose information if the value is outside the range of "int".

7.4 Floating-Point Precision

Floating-point numbers cannot represent every decimal value exactly. For example, "0.1 + 0.2" may not equal exactly "0.3". Such small errors can accumulate when many bills are added.

7.5 Rounding Rule

Money should be stored as integer paise to avoid floating-point rounding errors. If a calculation produces a fractional paise, it should be rounded to the nearest paisa at a clearly defined stage.

7.6 Alternative Concept and Complexity Analysis

Using floating-point calculations and rounding only at the end may introduce small errors across many bills. Integer-paise calculations are exact when integer overflow does not occur. Both approach

hes generally take constant time per bill and use constant memory per total.

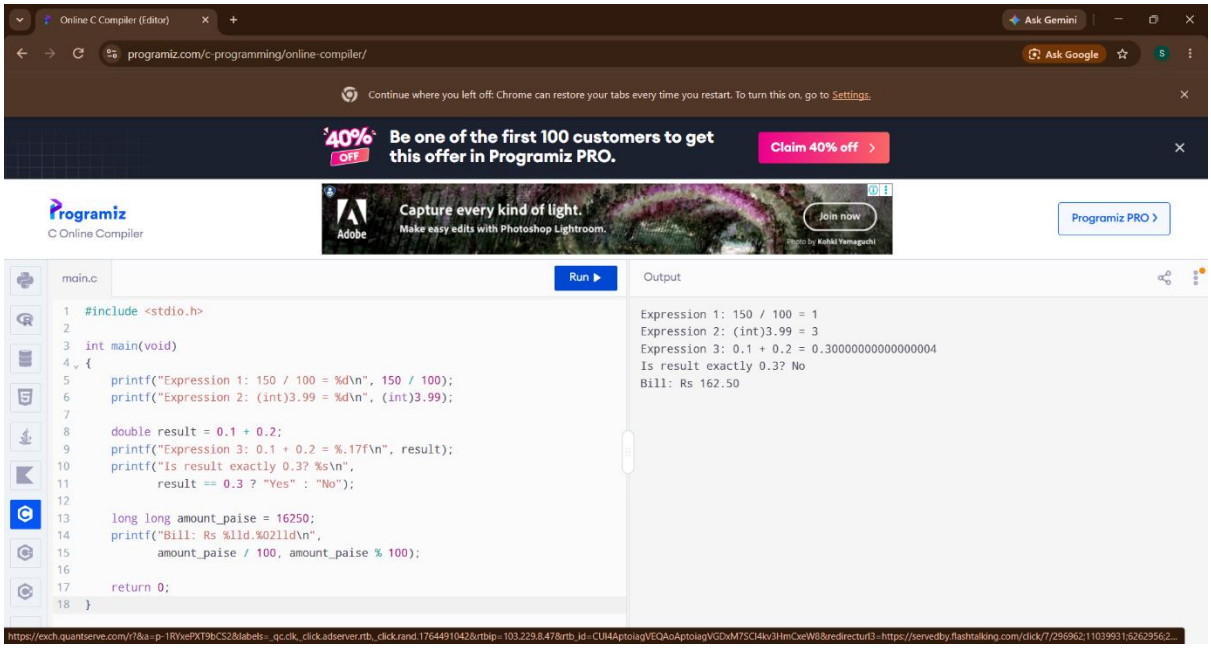
```
#include <stdio.h>


int main(void)
{
   printf("Expression 1: 150 / 100 = %d\n", 150 / 100);
   printf("Expression 2: (int)3.99 = %d\n", (int)3.99);


   double result = 0.1 + 0.2;
   printf("Expression 3: 0.1 + 0.2 = %.17f\n", result);
   printf("Is result exactly 0.3? %s\n",
         result == 0.3 ? "Yes" : "No");


   long long amount_paise = 16250;
   printf("Bill: Rs %lld.%02lld\n",
         amount_paise / 100, amount_paise % 100);


   return 0;
}
```


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main.c
Run
Output
#include <stdio.h>
int main(void)
{
printf("Expression 1: 150 / 100 = %d\n", 150 / 100);
printf("Expression 2: (int)3.99 = %d\n", (int)3.99);
double result = 0.1 + 0.2;
printf("Expression 3: 0.1 + 0.2 = %.17f\n", result);
printf("Is result exactly 0.3? %s\n",
result == 0.3 ? "Yes" : "No");
long long amount_paise = 16250;
printf("Bill: Rs %lld.%02lld\n",
amount_paise / 100, amount_paise % 100);
return 0;
}
Expression 1: 150 / 100 = 1
Expression 2: (int)3.99 = 3
Expression 3: 0.1 + 0.2 = 0.30000000000000004
Is result exactly 0.3? No
Bill: Rs 162.50